HEMSHANKAR LAUGI

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EDUCATION

B.S. (Honors) in Biomedical Engineering | University of Texas at Arlington (UTA) Graduation: May 2027
Minors: Mathematics, Physics **GPA: 3.96/4.00**
Honors Thesis: *Interpretable Machine Learning of Plasma Proteomics Reveals Stage-Specific Signatures Across the Alzheimer's Disease Continuum*
Thesis Advisor: Dr. Xi Zhu
Relevant Coursework: Calculus I-III, Linear Algebra & Differential Equations, Molecular Biology, Chemistry I-II, Physics I-II, Statistical Inference, Human Physiology, Linear Systems, Bioinformatics, Biomolecular Engineering, Medical Imaging, Bioinstrumentation, AI in Medicine, Digital Signals Processing

EXPERIENCE

Undergraduate Research Assistant | *Biomedical Research in AI & Neuroimaging (BRAIN) Lab, UTA* Oct 2025 – Present
- Led a plasma proteomics study from literature-driven identification of research gaps through computational analysis and biological interpretation.
- Designed a machine learning pipeline for multi-stage Alzheimer's classification using 146 plasma analytes from 566 ADNI participants.
- Evaluated 5 models with nested stratified 5×5 cross-validation and covariate adjustment, achieving stage-specific AUCs up to 0.963.
- Utilized SHAP interpretability to identify stage-specific proteomic signatures linked to immune, metabolic, inflammatory, and vascular pathways.
- Applied multivariable regression models to investigate protein-brain relationships by mapping plasma biomarkers to structural MRI measures.

Undergraduate Researcher | *NSF Research Training Group (RTG), Department of Mathematics, UTA* Sep 2025 – Present
- Co-developed an ordinary differential equation (ODE) model to simulate multidrug antibiotic resistance dynamics in *S. aureus* and *P. aeruginosa*.
- Estimated model parameters from experimental growth and antibiotic susceptibility data using nonlinear optimization in MATLAB.
- Synthesized literature to guide modeling of resistance through spontaneous mutation, back mutation, horizontal gene transfer, and fitness costs.
- Calibrated models to evaluate effects of multiple resistance mechanisms and treatment schedules on bacterial clearance and resistant populations.

Process Development Intern | *Amgen, Thousand Oaks, CA* May 2026 – Aug 2026
- Developed an automated cell culture scale-up workflow with 10X increase in volume throughput and reduction of up to 50% in manual resources.
- Optimized multi-plate sampling and passaging methods in an automated liquid handler, enabling 3X faster processing.
- Validated process accuracy, consistency, and no cross-contamination during automated passaging via automated cell counter and flow cytometry.
- Presented findings in weekly meetings, documented observations in electronic lab notebook (ELN), and prepared a job aid for workflow execution.

Analytical Science Intern | *Boehringer Ingelheim, Fremont, CA* May 2025 – Aug 2025
- Developed a highly reproducible MauriceFlex icIEF method that offers a 3X reduction in fractionation method development compared to CEX-HPLC.
- Achieved 45% reduction in process run time by deploying Supersonic icIEF method, streamlining protein characterization workflows.
- Improved vendor's peak prediction algorithm through empirical data analysis and collected charge fractions using 25% fewer reagents and time.
- Operated 17+ experiments to establish fractionation method, optimizing 10+ parameters to collect fractions with high purity and concentration.

Undergraduate Research Assistant | *Cancer Immunotherapy Lab, UTA* May 2024 – Apr 2025
- Conducted molecular, genetic, and protein studies involving nanobody development for immune checkpoint blockade, biochemical assays to understand interaction between TCR mimic antibodies and peptide MHC, and yeast surface display model for rapid antibody characterization.
- Contributed to a machine learning algorithm that predicts the binding specificity of nanobodies to cancer-specific peptide-MHC complexes.
- Documented experimental workflows and results to ensure data integrity, reproducibility, and efficient troubleshooting of experimental failures.

PROJECT

Handheld Facial Asymmetry Assessment Device | *Software, UI/UX & Computer Vision Lead* Aug 2026 – Present
- Developing a 3D computer vision pipeline using OAK-D depth imaging and point clouds for objective assessment of facial asymmetry.
- Implementing 3D facial landmark localization and XYZ coordinate extraction to quantify spatial differences in facial features.

MANUSCRIPTS

- **Laugi, H.**, Hosseinzadehketilateh, M., Lee, S., Weisberg, S. M., & Zhu, X. "Interpretable Machine Learning Links Group-Specific Plasma Proteomics to Regional Structural Neuroimaging in Alzheimer's Disease." *bioRxiv* Preprint, Nov 2026. DOI: 10.64898/2026.11.02.763368. Under Review at *Alzheimer's & Dementia: Diagnosis, Assessment & Disease Monitoring*.
- **Laugi, H.,** & Tang, H. "Transfer Learning for Alzheimer's MRI Classification: A Subject-Level Study of ImageNet and RadImageNet Pretraining." *Accepted**, 2026 IEEE BIBM Workshop on Machine Learning for Biological and Medical Image Big Data (ML4BMI).***
- Tang, H., & **Laugi, H.** "DenseNext: Parameter-Efficient Feature Generation through Fixed Spatial Aggregation in Densely Connected Networks." *Under Review.*

PRESENTATIONS

- **Laugi, H.,** et al. "*Mechanistic Modeling of Multidrug Antibiotic Resistance in Polymicrobial Coinfection."* **ABRCMS 2026**, Seattle, WA, Nov 2026.
- **Laugi, H.**, et al. *"From Plasma-Based Classification to Structural Neuroimaging Correlates Across the Alzheimer's Disease Continuum."* **University of Texas Systemwide Brain Research Summit 2026**, Austin, TX, Nov 2026.
- **Laugi, H.**, et al. *"Evaluating Multidrug Treatment Strategies for Polymicrobial Coinfection in Cystic Fibrosis."* **Tenth International Conference on Mathematical Modeling & Analysis of Populations in Biological Systems (ICMA-X)**, University of Maryland, College Park, MD, Oct 2026. Oral Presentation (30 min).
- **Laugi, H.**, Hosseinzadehketilateh, M., & Zhu, X. "*Interpretable Machine Learning of Plasma Proteomic Biomarkers for Multi-Stage Alzheimer's Disease Classification.*" **Honors Research Symposium**, University of Texas at Arlington, Arlington, TX, Apr 2026.
- Perry, I., **Laugi, H.**, Rosas, A., Dales, J., Huggans, H., & Kim, Y.-T. "*Neo-Limb: Adaptive Socket Intelligence for Improved Prosthetic Comfort.*" **College of Engineering Innovation Day**, University of Texas at Arlington, Arlington, TX, Apr 2026.
- **Laugi, H.**, & Mehra, R. "*Establishing Imaged Capillary Isoelectric Focusing (icIEF) Fractionation on MauriceFlex for High-Throughput Analytical Characterization of Protein Charge Variants.*" Summer Internship Presentation, **Boehringer Ingelheim**, Fremont, CA, Aug 2025.
- **Laugi, H.**, Ghaffari, S., & Weidanz, J. "*Engineering a Yeast Surface Display Platform for Peptide/MHC Expression and Rapid Characterization of T Cell Receptor Mimic (TCRm) Antibodies.*" **College of Engineering Innovation Day**, University of Texas at Arlington, Arlington, TX, Apr 2025.

SKILLS

- **Programming**: Python, MATLAB, R, Linux/Unix
- **Quantitative & Computational**: Statistical analysis, machine learning, deep learning, mathematical modeling, ordinary differential equations (ODEs), model interpretability, nonlinear optimization, multivariable regression
- **Biomedical Data Analysis**: Clinical, multi-omics, and neuroimaging data analysis
- **Experimental Methods**: Cell culture (suspension & adherent), flow cytometry, capillary isoelectric focusing (icIEF), western blotting, capillary electrophoresis, PCR, BCA assay, affinity chromatography, protein-binding assays, transfection, yeast surface display

AWARDS & GRANTS

- **Alfred R. and Janet H. Potvin Outstanding Bioengineering Student Award** | *UTA Bioengineering Department*, Apr 2025
- **Third Place, Undergraduate Research Presentation** | *UTA College of Engineering Innovation Day*; 57 undergraduate projects showcased, Apr 2025
- **Second Place, Research Presentation** | *TExBioMed 2024 Competition (15 interns)*; Full Sponsorship to Attend THIA 2024 Conference, Aug 2024
- **Honors College Special Opportunity Fund** | *UTA Honors College*; Conference Travel Support for ICMA-X 2026, Oct 2026
- **Carolyn A. Barros Founding Dean Endowed Scholarship** | *UTA Honors College*, 2026-2027
- **Dana Dunn Honors College Endowed Scholarship** | *UTA Honors College*, 2025-2026

LEADERSHIP & ADDITIONAL EXPERIENCE

Co-Founder & Vice-Chair | *IEEE Engineering in Medicine and Biology Society (EMBS), UTA Chapter* Jan 2026 – Present

- Co-founded UTA's first IEEE EMBS student chapter, securing organizational approval and building a 9-member interdisciplinary executive board.
- Lead technical and professional-development initiatives, including bioinformatics workshops and biomedical industry/lab tours for 20+ students.

Research Volunteer | *Cook Children's Neuroscience Research Center, Fort Worth, TX* Aug 2025 – Dec 2025

- Analyzed cerebral palsy data via DSI Studio tractography, focusing on diffusion MRI, fiber tract reconstruction, and structural connectivity analysis.
- Participated in patients' data collection sessions to understand various neuroimaging modalities and clinical data acquisition workflows.